NTA MOCK TEST

Q-1

Identify the wrong statement in context of heartwood.

(A) Organic compounds are deposited in it

(B) It is highly durable

(C) It conducts water and minerals efficiently

(D) It comprises dead elements with highly lignified walls

Q-2

Alexander Von Humboldt described for the first time:

(A) Ecological Biodiversity

(B) Laws of limiting factor

(C) Species area relationships

(D) Population Growth equation

Q-3

. Which of the following is made up of dead cells?

(A) Xylem parenchyma

(B) Collenchyma

(C) Phellem

(D) Phloem

NTA MOCK TEST

Q-4

The vascular cambium normally gives rise to

(A) Phelloderm

(B) Primary phloem

(C) Secondary xylem

(D) Periderm

NTA MOCK TEST

Q-5

In case of poriferans the spongocoel is lined with flagellated cells called

(A) Ostia

(B) Oscula

(C) Choanocytes

(D) Mesenchymal cells

Which among these is the correct combination of aquatic mammals?

(A) Seals, Dolphins, Sharks

(B) Dolphins, Seals, Trygon

(C) Whales, Dolphins Seals

(D) Trygon, Whales, Seals

Q-7

Which of the following represents the order of 'Horse'?

(A) Equidae

(B) Perissodactyla

(C) Caballus

(D) Ferus

NTA MOCK TEST

Q-8

Select the **correct** route for the passage of sperms in male frogs:

(A) Testes → Bidder's canal → Kidney → Vasa efferentia → Urinogenital duct → Cloaca

(B) Testes → Vasa efferentia → Kidney → Seminal Vesicle → Urinogenital duct → Cloaca

(C) Testes → Vasa efferentia → Bidder's canal → Ureter → Cloaca

(D) Testes → Vasa efferentia → Kidney → Bidder's canal → Urinogenital duct → Cloaca

NTA MOCK TEST

Q-9

Which one of the following is related to Ex-situ conservation of threatened animals and plants?

(A) Wildlife Safari parks

(B) Biodiversity hot spots

(C) Amazon rainforest

(D) Himalayan region

NTA MOCK TEST

Q-10

. An important characteristic that Hemichordates share with Chordates is

(A) Absence of notochord

(B) Ventral tubular nerve cord

(C) Pharynx with gill slits

(D) Pharynx without gill slits

NTA MOCK TEST

Q-11

. The region of Biosphere Reserve which is legally protected and where no human activity is allowed is known as :

(A) Core zone

(B) Buffer zone

(C) Transition zone

(D) Restoration zone

Viroids differ from viruses in having :

(A) DNA molecules with protein coat

(B) DNA molecules without protein coat

(C) RNA molecules with protein coat

(D) RNA molecules without protein coat

Which of the following are found in extreme saline conditions?

A) Archaebacteria

(B) Eubacteria

C) Cyanobacteria

(D) Mycobacteria

NTA MOCK TEST

Q-14

4. Mycorrhizae are the example of:

(A) Fungistasis

(B) Amensalism

(C) Antibiosis

(D) Mutualism

Q-15

. An example of colonial alga is

(A) Chlorella

(B) Volvox

(C) Ulothrix

(D) Spirogyra

. Spliceosomes are **not** found in cells of:

(A) Plants

(B) Fungi

(C) Animals

(D) Bacteria

Q-17

. Life cycle of *Ectocarpus* and *Fucus* respectively are

(A) Haplontic, Diplontic

(B) Diplontic, Haplodiplontic

(C) Haplodiplontic, Diplontic

(D) Haplodiplontic, Haplontic

NTA MOCK TEST

Q-18

Which among the following are the smallest living cells, known without a definite cell wall, pathogenic to plants as well as animals and can survive without oxygen?

(A) *Bacillus*

(B) *Pseudomonas*

(C) Mycoplasma

(D) *Nostoc*

DNA replication in bacteria occurs

(A) During s-phase

(B) Within nucleolus

(C) Prior to fission

(D) Just before transcription

Q-20

. Which of the following is not polymeric?

(A) Nucleic acids

(B) Proteins

(C) Polysaccharides

(D) Lipids

NTA MOCK TEST

Q-21

. Which one of the following statements is **correct**, with reference to enzymes?

(A) Apoenzyme = Holoenzyme + Coenzyme

(B) Holoenzyme = Apoenzyme + Coenzyme

(C) Coenzyme = Apoenzyme + Holoenzyme

(D) Holoenzyme = Coenzyme + Co-factor

Q-22

. A gene whose expression helps to identify transformed cell is known as

(A) Selectable marker

(B) Vector

(C) Plasmid

(D) Structural gene

. The association of histone H1 with a nucleosome indicates:

(A) Transcription is occurring

(B) DNA replication is occurring

(C) The DNA is condensed into a Chromatin Fibre

(D) The DNA double helix is exposed

NTA MOCK TEST

Q-24

. The DNA fragments separated on an agarose gel can be visualised after staining with:

(A) Bromophenol blue

(B) Acetocarmine

(C) Aniline blue

(D) Ethidium bromide

NTA MOCK TEST

Q-25

. What is the criterion for DNA fragments movement on agarose gel during gel electrophoresis?

(A) The larger the fragment size, the farther it moves

(B) The smaller the fragment size, the farther it moves

(C) Positively charged fragments move to farther end

(D) Negatively charged fragments do not move

NTA MOCK TEST

Q-26

. DNA fragments are

(A) Positively charged

(B) Negatively charged

(C) Neutral

(D) Either positively or negatively charged depending on their size

NTA MOCK TEST

Q-27

. The process of separation and purification of expressed protein before marketing is called

(A) Upstream processing

(B) Downstream processing

(C) Bioprocessing

(D) Postproduction processing

NTA MOCK TEST

Q-28

. The hepatic portal vein drains blood to the liver from

(A) Heart

(B) Stomach

(C) Kidney

(D) Intestine

29. Adult human RBCs are enucleate. Which of the following statement(s) is/are most appropriate explanation for this feature?

(i) They do not need to reproduce

(ii) They are somatic cells

(iii) They do not metabolize

(iv) All their internal space is available for oxygen transport

(A) Only (iv)

(B) Only (i)

(C) (i), (iii) and (iv)

(D) (ii) and (iii)

NTA MOCK TEST

Q-30

. MALT constitutes about _______ percent of the lymphoid tissue in human body

(A) 50%

(B) 20%

(C) 70%

(D) 10%

31. Lungs are made up of air-filled sacs the alveoli. They do not collapse even after forceful expiration, because of :

(A) Residual Volume

(B) Inspiratory Reserve Volume

(C) Tidal Volume

(D) Expiratory Reserve Volume

32. Attractants and rewards are required for:

(A) Anemophily

(B) Entomophily

(C) Hydrophily

(D) Cleistogamy

NTA MOCK TEST

Q-33

. Which of the following cell organelles is responsible for extracting energy from carbohydrates to form ATP?

(A) Lysosome

(B) Ribosome

(C) Chloroplast

(D) Mitochondrion

NTA MOCK TEST

Q-34

4. Which of the following components provides sticky character to the bacterial cell?

(A) Cell wall

(B) Nuclear membrane

(C) Plasma membrane

(D) Glycocalyx

NTA MOCK TEST

Q-35

. Which of the following options gives the correct sequence of events during mitosis?

(A) codensation → nuclear membrane disassembly → crossing over → segregation → telophase

(B) condensation → nuclear membrane disassembly → arrangement at equator → centromere division → segregation → telophase

(C) condensation → crossing over → nuclear membrane disassembly → segregation → telophase

(D) condensation → arrangement at equator → centromere division → segregation → telophase

NTA MOCK TEST

Q-36

. Zygotic meiosis is characterstic of

(A) *Marchantia*

(B) *Fucus*

(C) *Funaria*

(D) *Chlamydomonas*

NTA MOCK TEST

Q-37

37. Anaphase promoting complex (APC) is a protein degradation machinery necessary for proper mitosis of animal cells. If APC is defective in a human cell, which of the followin is expected to occur?

(A) Chromosomes will not condense

(B) Chromosomes will be fragmented

(C) Chromosomes will not segregate

(D) Recombination of chromosome arms will occur

. A decrease in blood pressure/volume will not cause the release of

(A) Renin

(B) Atrial Natriuretic Factor

(C) Aldosterone

(D) ADH

NTA MOCK TEST

Q-39

. GnRH, a hypothalamic hormone, needed in reproduction, acts on

(A) Anterior pituitary gland and stimulates secretion of LH and oxytocin

(B) Anterior pituitary gland and stimulates secretion of LH and FSH

(C) Posterior pituitary gland and stimulates secretion of oxytocin and FSH

(D) Posterior pituitary gland and stimulates secretion of LH and relaxin

NTA MOCK TEST

Q-40

. Hypersecretion of Growth Hormone in adults does not cause further increase in height, because

(A) Growth Hormone becomes inactive in adults

(B) Epiphyseal plates close after adolescence

(C) Bones loose their sensitivity to Growth Hormone in adults

(D) Muscle fibres do not grow in size after birth

NTA MOCK TEST

Q-41

. A temporary endocrine gland in the human body is

(A) Pineal gland

(B) Corpus cardiacum

(C) Corpus luteum

(D) Corpus allatum

NTA MOCK TEST

Q-42

42. Good vision depends on adequate intake of carotene rich food

Select the best option from the following statements

(i) Vitamin A derivatives are formed from carotene

(ii) The photopigments are embedded in the membrane discs of the inner segment

(iii) Retinal is a derivative of vitamin A

(iv) Retinal is a light absorbing part of all the visual photopigments

(A) (i) and (ii)

(B) (i), (iii) and (iv)

(C) (i) and (iii)

(D) (ii), (iii) and (iv)

NTA MOCK TEST

Q-43

. Which cells of 'Crypts of Lieberkuhn' secrete antibacterial lysozyme?

(A) Argentaffin cells

(B) Paneth cells

(C) Zymogen cells

(D) Kupffer cells

. A baby boy aged two years is admitted to play school and passes through a dental check-up. The dentist observed that the boy had twenty teeth. Which teeth were absent?

(A) Incisors

(B) Canines

(C) Pre-molars

(D) Molars

Which of the following options best represents the enzyme composition of pancreatic juice?

(A) Amylase, peptidase, trypsinogen, rennin

(B) Amylase, pepsin, trypsinogen, maltase

(C) Peptidase, amylase, pepsin, rennin

(D) Lipase, amylase, trypsinogen, procarboxypeptidase

NTA MOCK TEST

Q-46

. Which ecosystem has the maximum biomass?

(A) Forest ecosystem

(B) Grassland ecosystem

(C) Pond ecosystem

(D) Lake ecosystem

NTA MOCK TEST

Q-47

'. Presence of plants arranged into well-defined vertical layers depending on their height can be seen best in:

(A) Tropical Savannah

(B) Tropical Rain Forest

(C) Grassland

(D) Temperate Forest

NTA MOCK TEST

Q-48

. Which of the following in sewage treatment removes suspended solids?

(A) Tertiary treatment

(B) Secondary treatment

(C) Primary treatment

(D) Sludge treatment

). Which one of the following statements is not valid for aerosols?

(A) They are harmful to human health

(B) They alter rainfall and monsoon patterns

(C) They cause increased agricultural productivity

(D) They have negative impact on agricultural land

NTA MOCK TEST

Q-50

. Which of the following statements is correct?

(A) The ascending limb of loop of Henle is impermeable to water

(B) The descending limb of loop of Henle is impermeable to water

(C) The ascending limb of loop of Henle is permeable to water

(D) The descending limb of loop of Henle is permeable to electrolytes

Q-51

. Transplantation of tissues/organs fails often due to non-acceptance by the patient's body. Which type of immune-response is responsible for such rejections?

(A) Autoimmune response

(B) Cell-mediated immune response

(C) Hormonal immune response

(D) Physiological immune response

. Root hairs develop from the region of

(A) Maturation

(B) Elongation

(C) Root cap

(D) Meristematic activity

NTA MOCK TEST

Q-53

. Which of the following facilitates opening of stomatal aperture?

(A) Contraction of outer wall of guard cells

(B) Decrease in turgidity of guard cells

(C) Radial orientation of cellulose microfibrils in the cell wall of guard cells

(D) Longitudinal orientation of cellulose microfibrils in the cell wall of guard cells

. Capacitation occurs in

(A) Rete testis

(B) Epididymis

(C) Vas deferens

(D) Female Reproductive tract

NTA MOCK TEST

Q-55

. Capacitation occurs in

(A) Rete testis

(B) Epididymis

(C) Vas deferens

(D) Female Reproductive tract

NTA MOCK TEST

Q-56

. Out of 'X' pairs of ribs in humans only 'Y' pairs are true ribs. Select the option that correctly represents values of X and Y and provides their explanation:

(A)

$X = 12,\ Y = 7$	True ribs are attached dorsally to vertebral column and ventrally to the sternum

(B)

$X = 12,\ Y = 5$	True ribs are attached dorsally to vertebral column and sternum on the two ends

(C)

$X = 24,\ Y = 7$	True ribs are dorsally attached to vertebral column but are free on ventral side

(D)

$X = 24,\ Y = 12$	True ribs are dorsally attached to vertebral column but are free on ventral side

NTA MOCK TEST

Q-57

. Which of the following is correctly matched for the product produced by them?

(A) *Acetobacter aceti* : Antibiotics

(B) *Methanobacterium* : Lactic acid

(C) *Penicillium notatum* : Acetic acid

(D) *Saccharomyces cerevisiae* : Ethanol

58. If there are 999 bases in an RNA that codes for a protein with 333 amino acids, and the base at position 901 is deleted such that the length of the RNA becomes 998 bases, how many codons will be altered?

(A) 1

(B) 11

(C) 33

(D) 333

NTA MOCK TEST

Q-59

. The final proof for DNA as the genetic material came from the experiments of

(A) Griffith

(B) Hershey and Chase

(C) Avery, Mcleod and McCarty

(D) Hargobind Khorana

. Which of the following RNAs should be most adundant in animal cell?

(A) r-RNA

(B) t-RNA

(C) m-RNA

(D) mi-RNA

. During DNA replication, Okazaki fragments are used to enlongate

(A) The leading strand towards replication fork

(B) The lagging strand towards replication fork

(C) The leading strand away from replication fork

(D) The lagging strand away from the replication fork

. In *Bougainvillea* thorns are the modifications of

(A) Stipules

(B) Adventitious root

(C) Stem

(D) Leaf

NTA MOCK TEST

Q-63

. The morphological nature of the edible part of coconut is

(A) Perisperm

(B) Cotyledon

(C) Endosperm

(D) Pericarp

. Functional megaspore in an angiosperm develops into:

(A) Ovule

(B) Endosperm

(C) Embryo sac

(D) Embryo

Q-65

. Coconut fruit is a

(A) Drupe

(B) Berry

(C) Nut

(D) Capsule

NTA MOCK TEST

Q-66

66. Frog's heart when taken out of the body continues to beat for sometime

Select the best option from the following statements

(i) Frog is a poikilothermic

(ii) Frog does not have any coronary circulation

(iii) Heart is "myogenic" in nature

(iv) Heart is autoexcitable

(A) Only (iii)

(B) Only (iv)

(C) (i) and (ii)

(D) (iii) and (iv)

NTA MOCK TEST

Q-67

Myelin sheath is produced by

(A) Schwann Cells and Oligodendrocytes

(B) Astrocytes and Schwann Cells

(C) Oligodendrocytes and Osteoclasts

(D) Osteoclasts and Astrocytes

Q-68

. Receptor sites for neurotransmitters are present on

(A) Membranes of synaptic vesicles

(B) Pre-synaptic membrane

(C) Tips of axons

(D) Post-synaptic membrane

NTA MOCK TEST

Q-69

Asymptote in a logistic growth curve is obtained when

(A) The value of 'r' approaches zero

(B) K = N

(C) K > N

(D) K < N

Q-70

. Phosphoenolpyruvate (PEP) is the primary CO_2 acceptor in:

(A) C_3 plants

(B) C_4 plants

(C) C_2 plants

(D) C_3 and C_4 plants

Which statement is wrong for Krebs' cycle?

(A) There are three points in the cycle where NAD^+ is reduced to $NADH + H^+$

(B) There is one point in the cycle where FAD^+ is reduced to $FADH_2$

(C) During conversion of succinyl CoA to succinic acid, a molecule of GTP is synthesised

(D) The cycle starts with condensation of acetyl group (acetyl CoA) with pyruvic acid to yield citric acid

NTA MOCK TEST

Q-72

. With reference to factors affecting the rate of photosynthesis, which of the following statements is not correct?

(A) Light saturation for CO_2 fixation occurs at 10% of full sunlight

(B) Increasing atmospheric CO_2 concentration upto 0.05% can enhance CO_2 fixation rate

(C) C_3 plants responds to higher temperatures with enhanced photosynthesis while C_4 plants have much lower temperature optimum

(D) Tomato is a greenhouse crop which can be grown in CO_2 - enriched atmosphere for higher yield

Q-73

. Fruit and leaf drop at early stages can be prevented by the application of

(A) Cytokinins

(B) Ethylene

(C) Auxins

(D) Gibberellic acid

NTA MOCK TEST

Q-74

Select the mismatch :

(A) *Pinus* – *Dioecious*

(B) *Cycas* – *Dioecious*

(C) *Salvinia* – *Heterosporous*

(D) *Equisetum* – *Homosporous*

NTA MOCK TEST

Q-75

. Plants which produce characterstic pneumatophores and show vivipary belong to

(A) Mesophytes

(B) Halophytes

(C) Psammophytes

(D) Hydrophytes

. Select the mismatch:

(A) *Frankia – Alnus*

(B) *Rhodospirillum – Mycorrhiza*

(C) *Anabaena – Nitrogen fixer*

(D) *Rhizobium – Alfalfa*

. A disease caused by an autosomal primary non-disjunction is

(A) Down's syndrome

(B) Klinefelter's syndrome

(C) Turner's syndrome

(D) Sickle cell anemia

NTA MOCK TEST

Q-78

. Thalassemia and sickle cell anemia are caused due to a problem in globin molecule synthesis. Select the correct statement.

(A) Both are due to a qualitative defect in globin chain synthesis

(B) Both are due to a quantitative defect in globin chain synthesis

(C) Thalassemia is due to less synthesis of globin molecules

(D) Sickle cell anemia is due to a quantitative problem of globin molecules

NTA MOCK TEST

Q-79

. Which one from those given below is the period for Mendel's hybridization experiments?

(A) 1856 - 1863

(B) 1840 - 1850

(C) 1857 - 1869

(D) 1870 - 1877

. The genotypes of a Husband and Wife are $I^A I^B$ and $I^A i$. Among the blood types of their children, how many different genotypes and phenotypes are possible?

(A) 3 genotypes; 3 phenotypes

(B) 3 genotypes; 4 phenotypes

(C) 4 genotypes; 3 phenotypes

(D) 4 genotypes; 4 phenotypes

NTA MOCK TEST

Q-81

. Among the following characters, which one was not considered by Mendel in his experiments on pea?

(A) Stem-Tall or Dwarf

(B) Trichomes-Glandular or non-glandular

(C) Seed-Green or Yellow

(D) Pod-Inflated or Constricted

82. Match the following sexually transmitted diseases (Column - I) with their causative agent (Column - II) and select the correct option.

Column – I	Column – II
(a) Gonorrhea	(i) HIV
(b) Syphilis	(ii) Neisseria
(c) Genital Warts	(iii) Treponema
(d) AIDS	(iv) Human Papilloma-Virus

(A) (a) (b) (c) (d)
(ii) (iii) (iv) (i)

(B) (a) (b) (c) (d)
(iii) (iv) (i) (ii)

(C) (a) (b) (c) (d)
(iv) (ii) (iii) (i)

(D) (a) (b) (c) (d)
(iv) (iii) (ii) (i)

NTA MOCK TEST

Q-83

. The function of copper ions in copper releasing IUD's is:

(A) They suppress sperm motility and fertilizing capacity of sperms

(B) They inhibit gametogenesis

(C) They make uterus unsuitable for implantation

(D) They inhibit ovulation

NTA MOCK TEST

Q-84

. In case of a couple where the male is having a very low sperm count, which technique will be suitable for fertilisation?

(A) Intrauterine transfer

(B) Gamete intracytoplasmic fallopian transfer

(C) Artificial Insemination

(D) Intracytoplasmic sperm injection

NTA MOCK TEST

Q-85

. A dioecious flowering plant prevents both:

(A) Autogamy and xenogamy

(B) Autogamy and geitonogamy

(C) Geitonogamy and xenogamy

(D) Cleistogamy and xenogamy

. Flowers which have single ovule in the ovary and are packed into inflorescence are usually pollinated by

(A) Water

(B) Bee

(C) Wind

(D) Bat

NTA MOCK TEST

Q-87

. Double fertilization is exhibited by:

(A) Gymnosperms

(B) Algae

(C) Fungi

(D) Angiosperms

NTA MOCK TEST

Q-88

. Homozygous purelines in cattle can be obtained by:

(A) mating of related individuals of same breed

(B) mating of unrelated individuals of same breed

(C) mating of individuals of different breed

(D) mating of individuals of different species

. Artificial selection to obtain cows yielding higher milk output represents

(A) Stabilizing selection as it stabilizes this character in the population

(B) Directional as it pushes the mean of the character in one direction

(C) Disruptive as it splits the population into two one yielding higher output and the other lower output

(D) Stabilizing followed by disruptive as it stabilizes the population to produce higher yielding cows

. The water potential of pure water is

(A) Zero

(B) Less than zero

(C) More than zero but less than one

(D) More than one

Question No.	Correct Option
Question 1:	3
Question 2:	3
Question 3:	3
Question 4:	3
Question 5:	3
Question 6:	3
Question 7:	2
Question 8:	4
Question 9:	3
Question 10:	3
Question 11:	1
Question 12:	4
Question 13:	1
Question 14:	4
Question 15:	2

Question No.	Correct Option
Question 16:	4
Question 17:	3
Question 18:	3
Question 19:	3
Question 20:	4
Question 21:	2
Question 22:	1
Question 23:	3
Question 24:	4
Question 25:	2
Question 26:	2
Question 27:	2
Question 28:	4
Question 29:	1
Question 30:	1

Question No.	Correct Option
Question 31:	1
Question 32:	2
Question 33:	4
Question 34:	4
Question 35:	2
Question 36:	4
Question 37:	3
Question 38:	2
Question 39:	2
Question 40:	2
Question 41:	3
Question 42:	2
Question 43:	2
Question 44:	3
Question 45:	4

Question No.	Correct Option
Question 46:	1
Question 47:	2
Question 48:	3
Question 49:	3
Question 50:	1
Question 51:	2
Question 52:	1
Question 53:	3
Question 54:	4
Question 55:	3
Question 56:	1
Question 57:	4
Question 58:	3
Question 59:	2
Question 60:	1

Question No.	Correct Option
Question 61:	4
Question 62:	3
Question 63:	3
Question 64:	3
Question 65:	1
Question 66:	4
Question 67:	1
Question 68:	3
Question 69:	2
Question 70:	2
Question 71:	4
Question 72:	3
Question 73:	3
Question 74:	1
Question 75:	2

Question No.	Correct Option
Question 76:	2
Question 77:	1
Question 78:	3
Question 79:	1
Question 80:	3
Question 81:	2
Question 82:	1
Question 83:	1
Question 84:	3
Question 85:	2
Question 86:	3
Question 87:	4
Question 88:	1
Question 89:	2
Question 90:	1